



Optimizing ICU Mortality Prediction Models Using Class-Weighting & Threshold Tuning

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INTRODUCTION

- The eICU Collaborative Research Database contains ICU patient demographics, vitals, labs and clinical scores across 200+ U.S. hospitals
- ICU mortality prediction helps risk-stratify patients early, but real-world datasets are highly imbalanced (fewer deaths, many survivors)
- This imbalance affects model learning, threshold choice, and fairness across subgroups.

Dataset : eICU-CRD v2.0 (demo) from PhysioNet
Outcome: actual ICU mortality (0 = survived, 1 = died)

OBJECTIVE

Can we optimize an ICU mortality prediction model by using class-weighted training and threshold tuning, and what trade-offs does this create in recall, precision, and fairness across subgroups?

METHODOLOGY

Data Preprocessing:

- Merged tables and built clean 24-hr vital/labs view (min/mean/max)
- Removed impossible values (MAP<10, SpO₂<50, HR=0)
- Log-transformed skewed labs; winsorized outliers
- Imputed missing values by ICU type (median)
- Encoded categorical variables

Modeling Workflow:

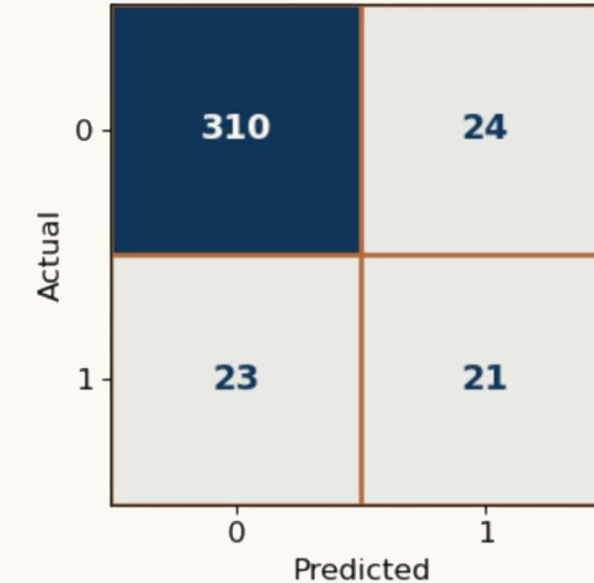
- Split 70/15/15 (train/val/test)
- Built Random Forest and XGBoost with class_weight=balanced
- Tuned probability thresholds to maximize F1 (balance recall/precision)
- Evaluated subgroup fairness (gender, ethnicity, ICU type)

RESULTS & KEY FINDINGS

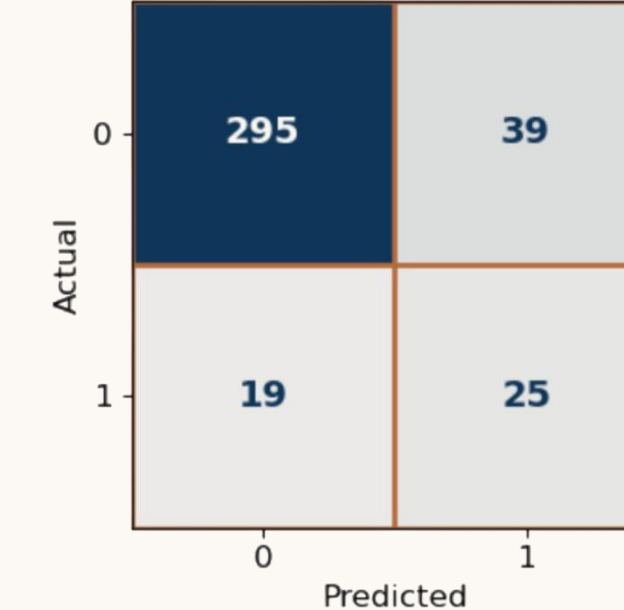
Model Performance

XGBoost					Random Forest				
XGB TEST report (balanced threshold):					RF TEST report (balanced threshold):				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.931	0.928	0.930	334	0	0.939	0.883	0.910	334
1	0.467	0.477	0.472	44	1	0.391	0.568	0.463	44
accuracy	0.876			378	accuracy	0.847			378

XGB Test Confusion Matrix (t=0.36)

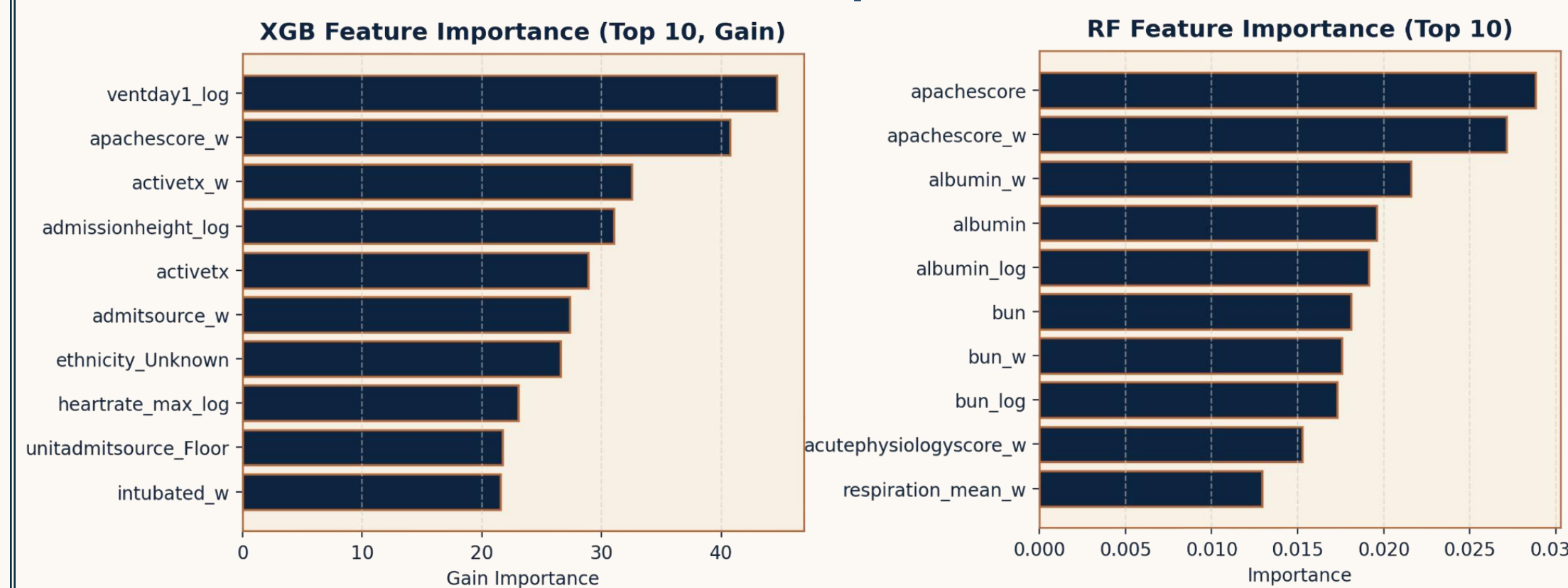


RF Test Confusion Matrix (t=0.25)



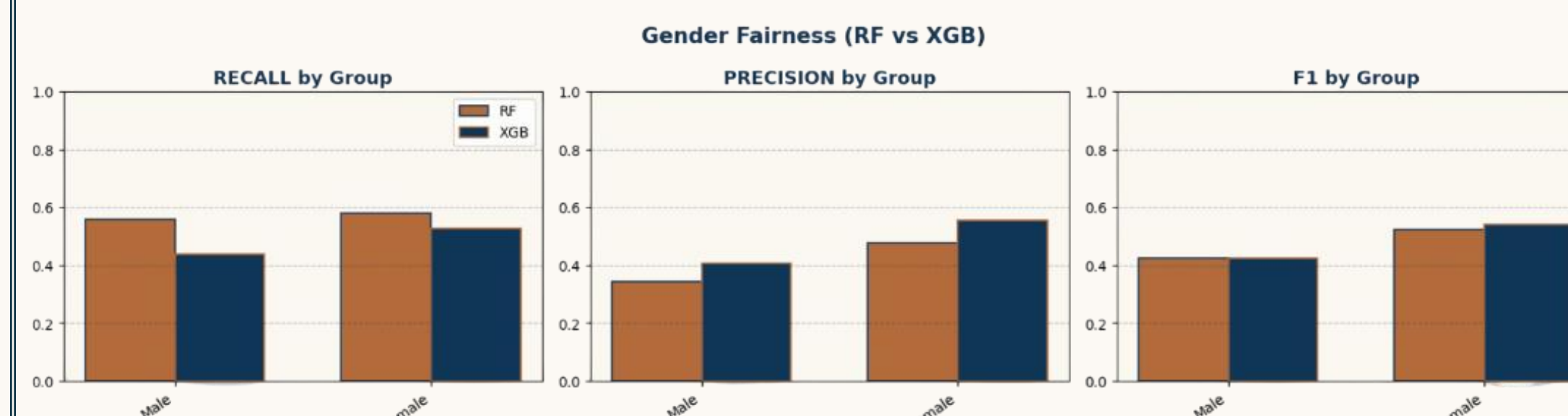
- Threshold tuning improved minority-class recall for both models

Feature Importance



- Random Forest focuses on physiology; XGBoost adds early interventions
- APACHE, albumin, and renal markers dominate both of the models

Fairness Evaluation



- Fairness evaluations showed no major model-driven disparities

CONCLUSIONS

- Class-weighted Random Forest and XGBoost improved mortality detection
- XGBoost achieved slightly higher accuracy while Random Forest offered more balanced recall/precision.
- Top predictors were consistent with clinical severity markers (APACHE, albumin, BUN, ventilation).
- Fairness checks showed no major disparities; variation was mostly due to small subgroup sizes.

CHALLENGES & IMPROVEMENTS

- Class Imbalance: Used class-weights and threshold tuning to detect rare deaths without overfitting
- Sparse subgroups: Some fairness metrics were unstable due to very small sample sizes.
- Skewed clinical features: Vitals and lab values required log-transform/winsorizing for stable modeling

REFERENCES

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<https://www.nature.com/articles/s41598-020-78184-7>
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